BraiNCA: brain-inspired neural cellular automata and applications to morphogenesis and motor control

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Motivation: from NCA to BraiNCA

Most of the Neural Cellular Automata (NCAs) defined in the literature have a common theme: they are based on regular grids with a Moore neighborhood (one-hop neighbors) [1, 2]. **Vanilla NCAs don't take into account long-range connections and more complex topologies such as those found in the brain (A).**

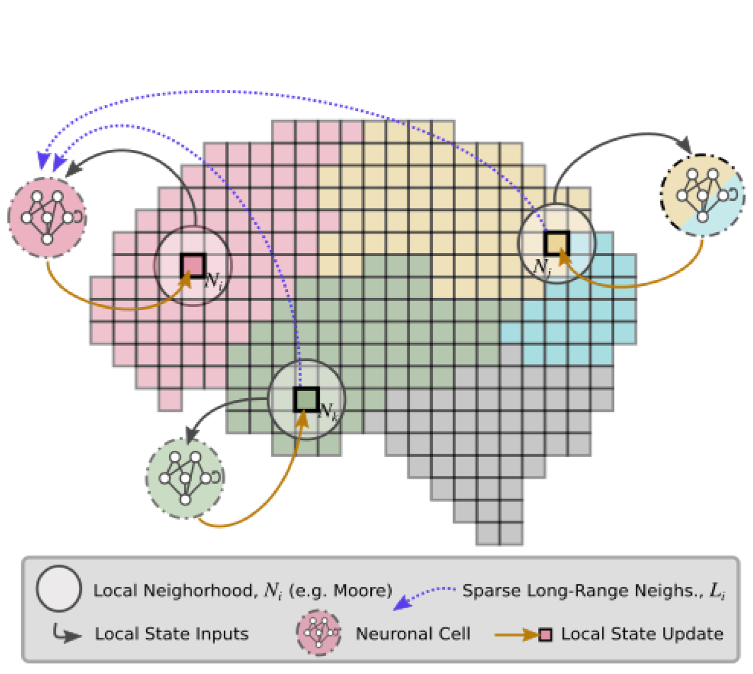
BraiNCA is a brain-inspired NCA with an attention layer, long-range connections and custom (non-square) topologies:

- From purely local aggregation
- To attention-weighted neighborhood integration [3],
- New topologies and spatial-functional relationships.

Each node **dynamically gates information** from its neighbors. By explicitly modelling **long-range interactions**, we augment the underlying connectivity to enable information flow beyond the one-hop neighborhoods used by Vanilla NCAs.

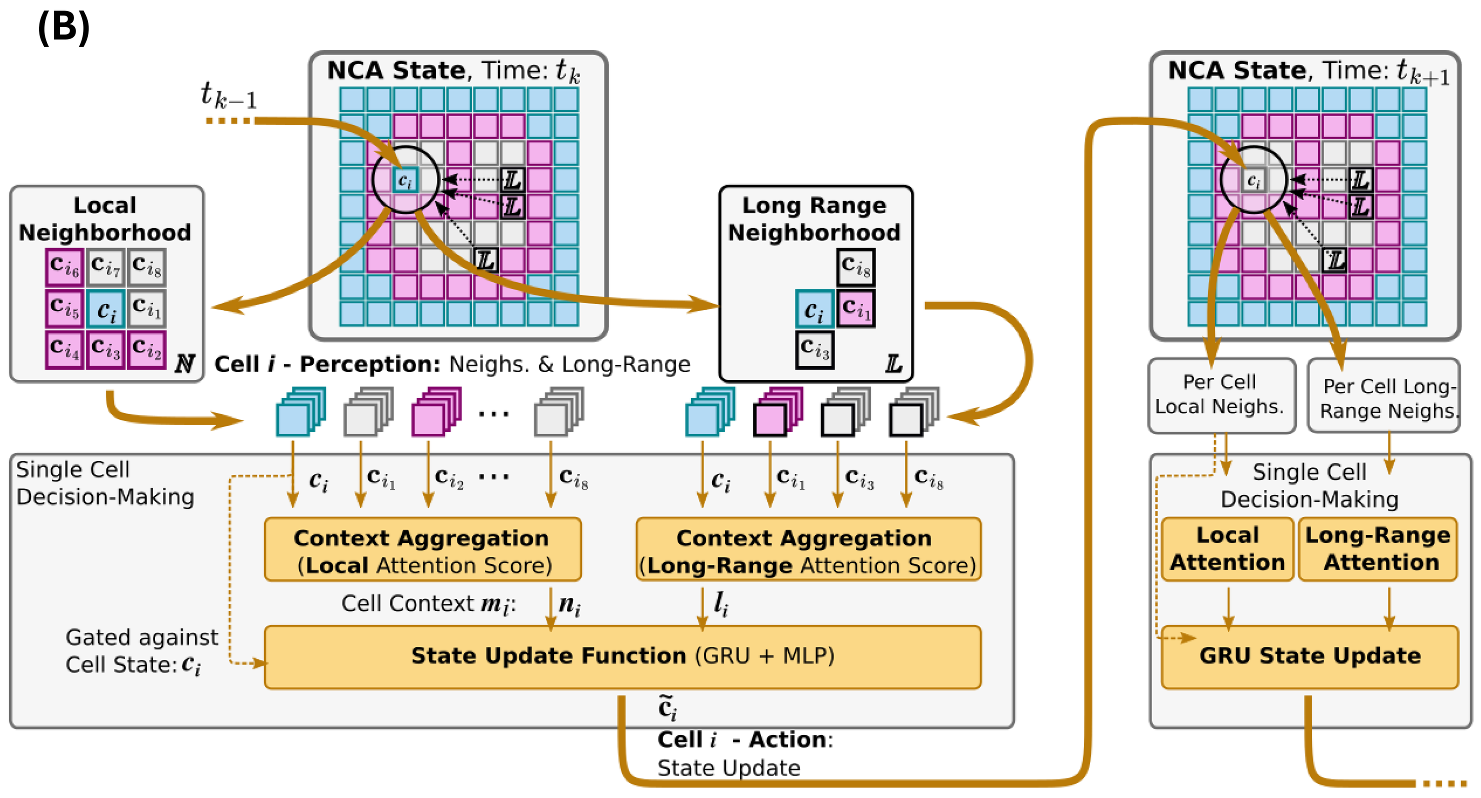
BraiNCA shows better results in terms of robustness and speed of learning on both a morphogenetic and a control task.

(A)



Schematics of the BraiNCA model. Every cell not only integrates its local neighborhood N_i via attention, but also long-range signals from sparsely connected distant cells j, $k \in L_i$ for recurrent state updates. Flexible topologies are allowed, departing from NCAs with regular, e.g., square grid-layouts.

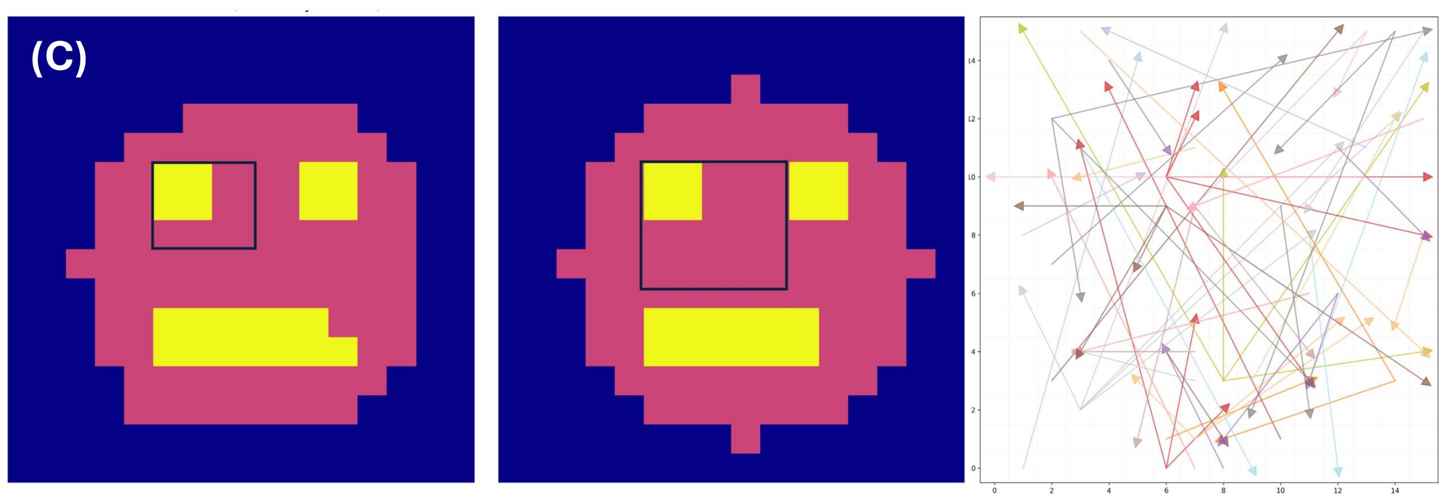
(B)



System-Level Control via Decentralized Decision-Making. Each cell's state is updated by aggregating information from both local and long-range neighborhoods through attention-based context encoding. The combined signals are processed via a recurrent update function (GRU + MLP), enabling spatially distributed coordination in morphogenesis experiments and sensory-motor control of RL agents.

Morphogenetic task

(C)



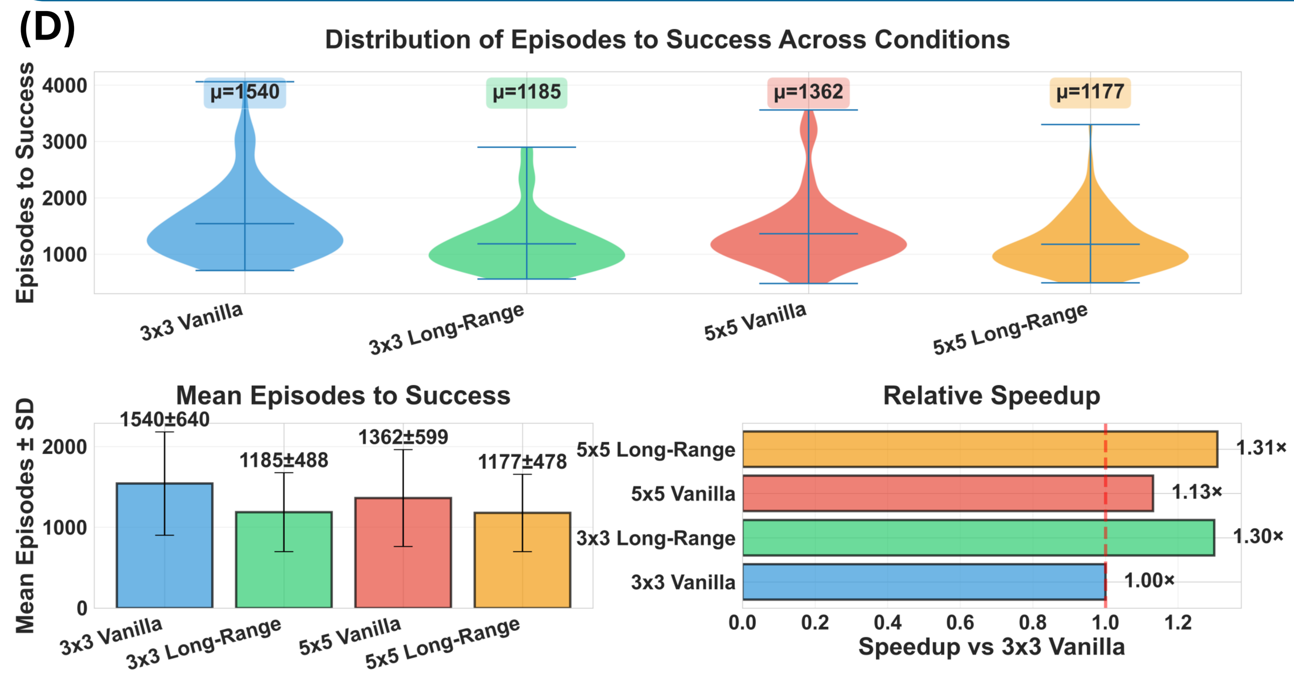
3x3 neighborhood | 5x5 neighborhood | Example of long-range connections

Morphogenesis task. Left: target pattern and cell labeling for the 3×3 neighborhood condition. Middle: target pattern with a 5×5 neighborhood. Right: example of sparse long-range connections used in long-range variants, mimicking scale-free networks found in the brain.

Contrast purely local (Vanilla NCA) and long-range connectivity (BraiNCA) on pattern formation task of 16×16 smiley-face (C).
Success: 98% or greater pixel-wise accuracy with the target pattern after **fixed developmental period of T = 35 cellular update steps**.
Training: **gradient descent** on network parameters minimizes deviation of achieved from target patterns over repeated developmental episodes starting from **fixed random initialization**.

Long-range connections and neighborhood size increase accuracy and speed of learning on the morphogenetic task

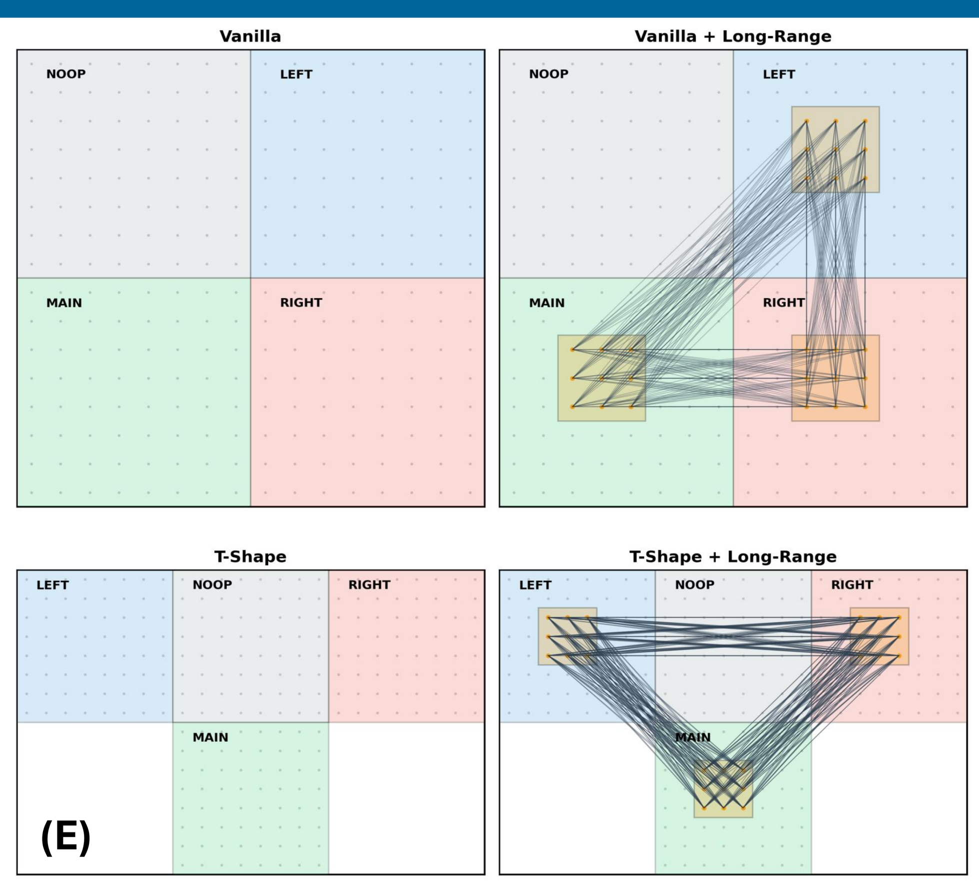
(D)



Long-range connections and size of neighborhood integration increase accuracy and speed of learning. Top: Violin plots of the episodes to success in the different conditions. Bottom left: Mean episodes to success in the different conditions. Bottom right: Relative speedup compared to the baseline (3×3 Vanilla)

- 3×3 Long-Range (BraiNCA) vs. 3×3 Vanilla (baseline): **long-range improves training speed by 30%**
- 5×5 Vanilla vs. 3×3 Vanilla: **larger neighborhood improves training speed by 13%**
- 5×5 Long-Range (best) vs. 3×3 Vanilla: **long-range & neighborhood improve training speed by 31%**
- 5×5 Long-Range vs. 5x5 Vanilla: **sparse long-range connections beat larger, dense neighborhoods**

Lunar Lander task



(E)

LunarLander connectivity schematics for action selection: Cells vote locally; regional activity is pooled into four logits (NOOP/LEFT/MAIN/RIGHT); action is sampled via softmax.

Vanilla: 16×16 grid with 8×8 quadrant action regions

Vanilla+LR: Vanilla grid and additional long-range cross-zone connectivity for messaging between yellow patches

T-Shape: 24×16 grid with four active 8×8 zones in a T-shape

T-Shape+LR: T-Shape topology and cross-zone messaging

Somatotopic topology increases robustness and speed of learning

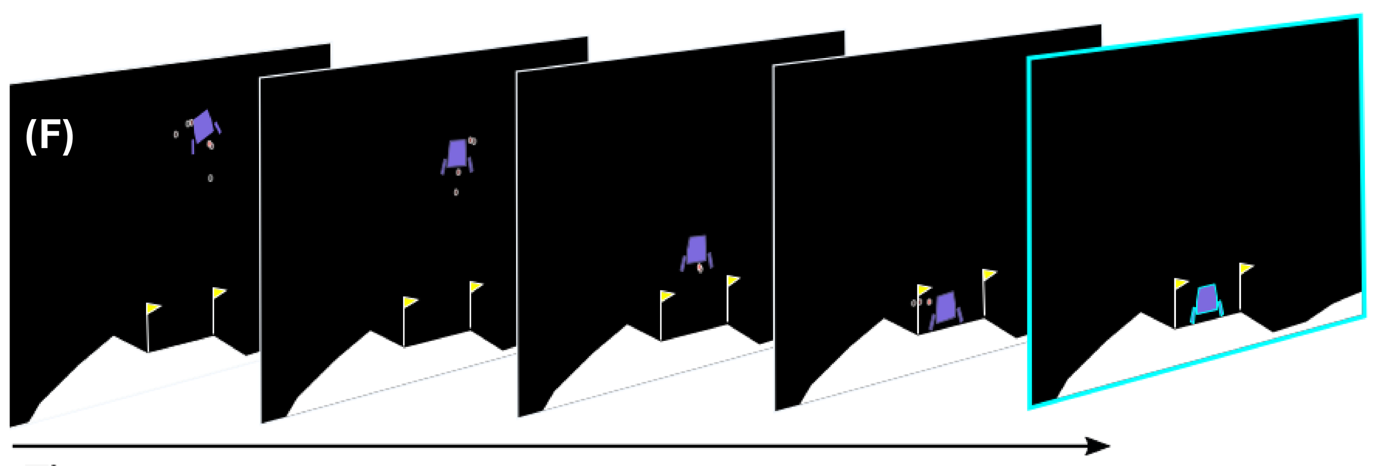
Task-aligned motor-cortex somatotopy [4] via T-Shape topology:
Four action zones (LEFT / NOOP / RIGHT / MAIN) **align NCA geometry** with actuator function, i.e., with the **lander's spatial-functional setting**.

Increased speed of learning and **robustness** compared to Vanilla**:**
+13.8 pp Success Rate and **-1121 RMST** (episodes to success)

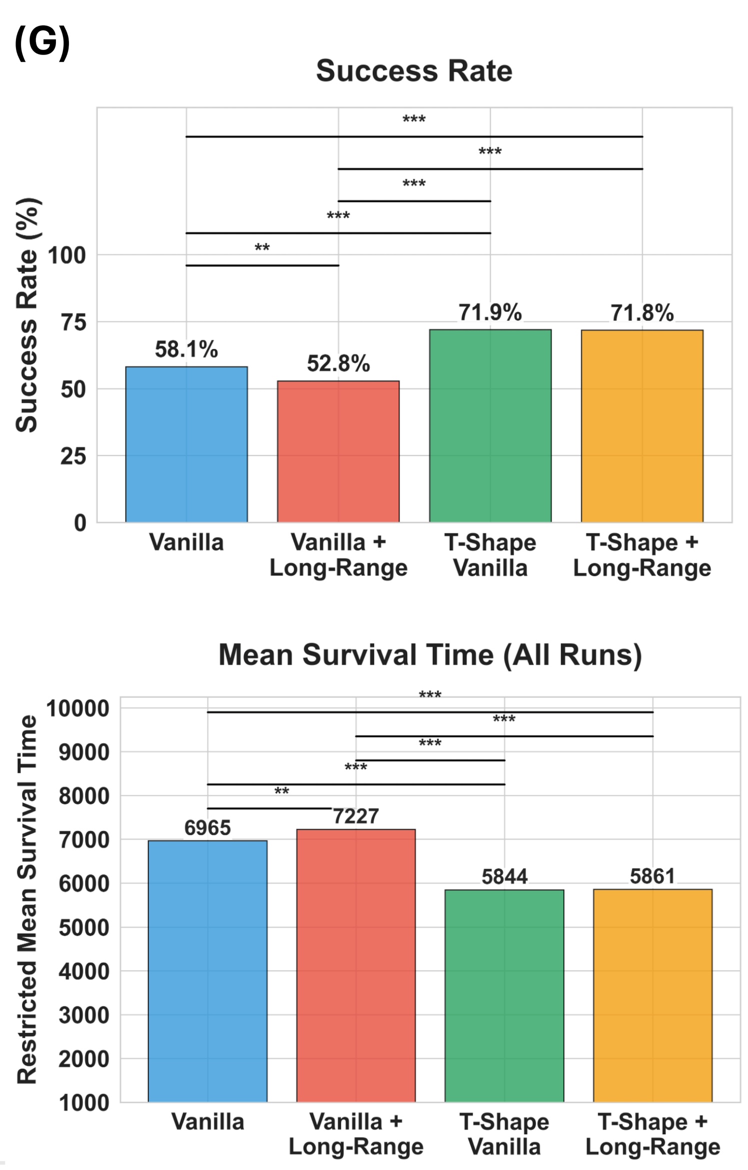
Notably, adding the current long-range cross-zone connections **slightly reduces robustness and learning speed** within both topologies.

Future studies will explore **learning optimal BraiNCA connectivity**.

(F)



(G)



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